Within the flat 6D Conformal Spacetime Algebra system ($Cl(4,1,1)$), we can construct a complex vector space to represent local realist probability theory.

In this representation, points in the space correspond to **propositions** about the physical system. Probabilities are weighted by complex amplitudes, where the modulus squared of the amplitude corresponds to the classical probability that a proposition is true, and the phase corresponds to the physical precessional phase of the local soliton current loops [13.2] associated with that proposition.

Section 14: Construction of the Complex Vector Space for Probability Theory

This section mathematically formalizes a probabilistic state space, where propositions about a system are encoded as complex vectors, enabling the computation of expectations through quadratic forms.

14.1 Complex Vector Space of Probability Amplitudes for Propositions
We construct a complex vector space $\mathcal{H} \cong \mathbb{C}^N$ where each basis vector $|k\rangle$ represents a **mutually exclusive elementary proposition** about the system's physical state or the outcome of a measurement (e.g., "The Josephson junction is in state [0]," or "The SQUID detects clockwise current").

Any composite proposition, or a complete state of knowledge about the system, is represented as a linear combination of these elementary propositions:

$$|\Psi\rangle = \sum_{k=1}^{N} c_k |k\rangle$$

where $c_k \in \mathbb{C}$ are the complex probability weights (amplitudes) associated with each proposition. Each complex weight can be written in polar form:

$$c_k = \sqrt{p_k} e^{i\varphi_k}$$

In this context: * $p_k = |c_k|^2 \in [0,1]$ represents the classical probability that the elementary proposition k is true. * φ_k represents the physical precessional phase of the local soliton current loops [13.2] that physically underpin the truth value of proposition k. This phase adds a richer structure to probability, allowing for interference-like effects (e.g., when propositions about distinct paths interfere).

The completeness axiom of probability requires that the sum of probabilities for all mutually exclusive elementary propositions is unity:

$$\sum_{k=1}^{N} |c_k|^2 = 1$$

14.2 Tensor Space Representation: Column and Row Matrices for Propositions The proposition vector $|\Psi\rangle$ is represented as a column matrix $\mathbf{\Psi}$ in this complex tensor space:

$$\mathbf{\Psi} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{pmatrix}$$

The dual space of this vector space consists of row matrices. The dual proposition vector $\langle\Psi|$ is represented by the conjugate transpose (adjoint) of the column matrix $\mathbf{\Psi}$:

$$\mathbf{\Psi}^\dagger = \begin{pmatrix} c_1^* & c_2^* & \cdots & c_N^* \end{pmatrix}$$

The inner product of two sets of propositions, represented by $|\Phi\rangle = \sum a_k|k\rangle$ and $|\Psi\rangle = \sum c_k|k\rangle$, is the matrix product of the dual row and the column matrices:

$$\langle\Phi|\Psi\rangle = \mathbf{\Phi}^\dagger\mathbf{\Psi} = \begin{pmatrix} a_1^* & a_2^* & \cdots & a_N^* \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{pmatrix} = \sum_{k=1}^{N} a_k^* c_k$$

This inner product quantifies the probabilistic overlap or compatibility between two distinct sets of propositions about the system.

14.3 Quadratic Forms for Expectations of Propositional Observables Let $\mathbf{M}$ be an $N \times N$ square matrix representing a physical observable whose value depends on the truth of the elementary propositions. $\mathbf{M}$ can be thought of as a linear operator that maps propositions to propositions.

We construct a quadratic form by sandwiching the matrix $\mathbf{M}$ between a dual row vector $\mathbf{\Psi}^\dagger$ and a column vector $\mathbf{\Psi}$:

$$\langle\mathbf{M}\rangle = \mathbf{\Psi}^\dagger\mathbf{M}\mathbf{\Psi}$$

This quadratic form yields a real scalar that represents the **expectation** of the physical observable $\mathbf{M}$, given the probabilities and phases encoded in the proposition vector $\mathbf{\Psi}$. For example, this could compute the expectation of the voltage from a SQUID readout [13.4] or the correlation coefficient in an EPR-Bohm test [9.5].

Expanding this matrix sandwich product yields:

$$\langle \mathbf{M} \rangle = \begin{pmatrix} c_1^* & c_2^* & \cdots & c_N^* \end{pmatrix} \begin{pmatrix} M_{11} & M_{12} & \cdots & M_{1N} \\ M_{21} & M_{22} & \cdots & M_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ M_{N1} & M_{N2} & \cdots & M_{NN} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{pmatrix}$$

$$\langle \mathbf{M} \rangle = \sum_{i=1}^{N} \sum_{j=1}^{N} c_i^* M_{ij} c_j$$

This sandwich structure $\mathbf{\Psi}^\dagger \mathbf{M} \mathbf{\Psi}$ mathematically represents the complete algebraic projection of the system's precessional phase states onto the physical measuring device's coupling matrix $\mathbf{M}$. This allows for the computation of expectations for complex propositional systems, recovering the identical expectations of quantum probability using entirely classical wave mechanics and local precessional phases, but framed as operations on a rich probability space of propositions.